

# Metric regularity of the coordinate map for one-dimensional order isomorphisms

New partial result for the questions in arXiv:1408.4930

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**Status.** **New partial result; pointwise differentiability remains open.** For order isomorphisms between differentiable function spaces on real intervals, the associated homeomorphism cannot have a cusp, a flat point, or an infinite metric slope. It and its inverse are locally absolutely continuous, and its derivative is finite and nonzero almost everywhere.

## Source question

Leung and Tang [1] ask whether the homeomorphism associated to an order isomorphism  $C^p(X) \rightarrow C^p(Y)$  must be differentiable, or even  $C^p$ . They emphasize that this is unknown already for  $p = 1$  and  $X = Y = \mathbb{R}$ , and separately ask whether different smoothness orders can be order-isomorphic.

NONLINEAR ORDER ISOMORPHISMS ON FUNCTION SPACES

5

The final section, Section 8, concludes with some results characterizing order isomorphisms between spaces of the same type. At the end of the paper, we append a table summarizing the comparison results obtained. It is worth pointing out that the remaining open cases all concern spaces of differentiable functions. To take the space  $C^p(X)$  as an example, we have the following open problems. Suppose that

- (a) Suppose that  $T : C^p(X) \rightarrow C^p(Y)$  is an order isomorphism. Must the associated homeomorphism be differentiable on  $X$ ?  $C^p$  on  $X$ ? To illustrate the extent of our ignorance, the answers are unknown even for  $p = 1$  and  $X = Y = \mathbb{R}$ .
- (b) Is it possible to have  $p \neq q$  and some open sets  $X$  and  $Y$  in Banach spaces so that  $C^p(X)$  is order isomorphic to  $C^q(Y)$ ? (Here, we ask that  $C^p(X)$  contains a bump function.)

### 1. GENERAL FRAMEWORK

*Source evidence:* arXiv:1408.4930, p. 5.

Let  $I, J \subset \mathbb{R}$  be nonempty open intervals and let  $p, q \in \mathbb{N} \cup \{\infty\}$ ,  $p, q \geq 1$ . The representation theorem of [1] associates to every order isomorphism  $T : C^p(I) \rightarrow C^q(J)$  a homeomorphism  $\varphi : I \rightarrow J$  and increasing homeomorphisms  $\Phi(y, \cdot) : \mathbb{R} \rightarrow \mathbb{R}$  such that

$$Tf(y) = \Phi(y, f(\varphi^{-1}(y))). \quad (1)$$

For fixed  $t$ , the function  $y \mapsto \Phi(y, t) = T\bar{t}(y)$  lies in  $C^q(J)$ , where  $\bar{t}$  denotes the constant function.

## Idea of the proof

Suppose  $\varphi$  flattened a sequence: its output increments  $\delta_n$  would be negligible compared with input increments  $\varepsilon_n$ . Transforming the affine pencil  $c \mapsto c \pm (x - x_0)$  says that each vertical homeomorphism  $H_n(t) = \Phi(\varphi(x_n), t)$  maps every scalar interval of length  $\varepsilon_n$  to length  $O_c(\delta_n)$ . Baire category makes the constant uniform on a whole scalar interval. Partitioning that interval into steps of length  $\varepsilon_n$  would collapse a fixed positive vertical distance to zero, contradicting monotonicity and convergence of the constant images. Applying the same argument to  $T^{-1}$  gives the upper metric bound. A countable Lipschitz-piece argument and Banach–Zarecki then give absolute continuity in both directions.

**Lemma 1** (Baire partition obstruction). *For every  $x_0 \in I$ ,*

$$\liminf_{x \rightarrow x_0} \frac{|\varphi(x) - \varphi(x_0)|}{|x - x_0|} > 0.$$

*Proof.* Assume otherwise. There are  $x_n \rightarrow x_0$ ,  $x_n \neq x_0$ , such that, with

$$\varepsilon_n = |x_n - x_0|, \quad \delta_n = |\varphi(x_n) - \varphi(x_0)|,$$

one has  $\delta_n/\varepsilon_n \rightarrow 0$ . Pass to a subsequence on one side of  $x_0$ , and choose  $\sigma \in \{-1, 1\}$  so that  $\sigma(x_n - x_0) = \varepsilon_n$ . Put

$$y_n = \varphi(x_n), \quad y_0 = \varphi(x_0), \quad H_n(t) = \Phi(y_n, t).$$

For every  $c \in \mathbb{R}$ , take the affine function  $f_c(x) = c + \sigma(x - x_0)$ . Formula (1) gives

$$H_n(c + \varepsilon_n) - H_n(c) = T f_c(y_n) - T \bar{c}(y_n). \quad (2)$$

The function  $T f_c - T \bar{c}$  belongs to  $C^1(J)$  and vanishes at  $y_0$ . Therefore, for each fixed  $c$ , there are integers  $M, N$  such that

$$0 \leq H_n(c + \varepsilon_n) - H_n(c) \leq M \delta_n \quad (n \geq N). \quad (3)$$

Fix a compact scalar interval  $K$  with nonempty interior. For positive integers  $M, N$ , let  $E_{M,N} \subset K$  be the set of all  $c$  for which (3) holds for every  $n \geq N$ . Each  $E_{M,N}$  is closed, because every  $H_n$  is continuous, and their countable union is  $K$ . Baire's theorem supplies  $M, N$  and a nondegenerate interval  $[a, b] \subset E_{M,N}$ .

Let  $r_n = \lfloor (b - a)/\varepsilon_n \rfloor$ . Since  $H_n$  is increasing, partitioning from  $a$  into  $\varepsilon_n$ -steps and allowing one final overshoot gives, for all large  $n$ ,

$$\begin{aligned} 0 < H_n(b) - H_n(a) &\leq \sum_{j=0}^{r_n} [H_n(a + (j+1)\varepsilon_n) - H_n(a + j\varepsilon_n)] \\ &\leq M(r_n + 1)\delta_n \\ &\leq M \left( \frac{b-a}{\varepsilon_n} + 1 \right) \delta_n \longrightarrow 0. \end{aligned}$$

On the other hand, for fixed  $t$ ,  $H_n(t) = T \bar{t}(y_n) \rightarrow T \bar{t}(y_0) = H_0(t)$ . Hence the left side converges to  $H_0(b) - H_0(a) > 0$ , because  $H_0$  is an increasing homeomorphism. This contradiction proves the lemma.  $\square$

**Theorem 2** (pointwise metric regularity). *For every  $x_0 \in I$ ,*

$$0 < \liminf_{x \rightarrow x_0} \frac{|\varphi(x) - \varphi(x_0)|}{|x - x_0|} \leq \limsup_{x \rightarrow x_0} \frac{|\varphi(x) - \varphi(x_0)|}{|x - x_0|} < \infty.$$

*Proof.* Lemma 1 is the first inequality. Apply the same lemma to the order isomorphism  $T^{-1} : C^q(J) \rightarrow C^p(I)$ , whose associated homeomorphism is  $\varphi^{-1}$ . Thus, near  $y_0 = \varphi(x_0)$ ,

$$|\varphi^{-1}(y) - x_0| \geq c|y - y_0|$$

for some  $c > 0$ . Substituting  $y = \varphi(x)$  gives the finite upper bound.  $\square$

**Corollary 3** (bi-absolute continuity). *The restrictions of  $\varphi$  and  $\varphi^{-1}$  to compact subintervals are absolutely continuous. Moreover,*

$$0 < |\varphi'(x)| < \infty \quad \text{for almost every } x \in I.$$

*Proof.* Fix a compact interval  $L \Subset I$ . The finite upper metric slope in Theorem 2 covers  $L$  by countably many sets  $E_{m,n}$  on which

$$|\varphi(z) - \varphi(x)| \leq m|z - x| \quad \text{whenever } x, z \in L, |z - x| < 1/n.$$

After subdividing  $L$  into finitely many intervals of length less than  $1/n$ ,  $\varphi$  is  $m$ -Lipschitz on each corresponding piece of  $E_{m,n}$ . Hence  $\varphi$  maps every null subset of  $L$  to a null set. A real homeomorphism is monotone; the Banach–Zarecki theorem [2] now implies that  $\varphi|_L$  is absolutely continuous. The same argument applies to  $\varphi^{-1}$ .

Absolute continuity gives a finite derivative almost everywhere. If  $Z = \{x : \varphi'(x) = 0\}$  had positive measure, absolute continuity would give measure zero to  $\varphi(Z)$ . The null-set property of  $\varphi^{-1}$  would then give measure zero to  $Z = \varphi^{-1}(\varphi(Z))$ , a contradiction. Thus the derivative is nonzero almost everywhere.  $\square$

## Scope and full-upgrade audit

| Route                                      | Outcome                                                                                                               |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Cusp or singular coordinate counterexample | Excluded by the Baire partition obstruction and its inverse.                                                          |
| Pointwise metric slope                     | Positive lower and finite upper slope at every point.                                                                 |
| Measure-theoretic upgrade                  | Both coordinate directions are locally absolutely continuous; the derivative is nonzero and finite almost everywhere. |
| Affine-pencil tangent limits               | Blocked by the possible singularity of the vertical homeomorphism $t \mapsto \Phi(y_0, t)$ .                          |
| Higher-order nonnegative contact           | Gives additional small-o estimates but no unique first-order tangent for $\varphi$ .                                  |
| Different smoothness orders                | The theorem works for arbitrary $p, q \geq 1$ , but does not rule out $p \neq q$ .                                    |
| Jointly $C^1$ vertical factor              | Would imply local $C^1$ regularity by the implicit function theorem, but is not forced by the source hypotheses.      |

Thus the source’s pointwise differentiability,  $C^p$ -regularity, and distinct-order questions remain open. The result nevertheless rules out the entire singular-homeomorphism and cusp classes as possible associated maps.

**Novelty and verification.** Exact-title, exact-question,  $C^1(\mathbb{R})$ , metric-slope, bi-absolute-continuity, and later order-isomorphism searches through 11 August 2026 found no statement of Theorem 2 or Corollary 3. Human review should focus on the Baire uniformization, the final overshooting interval in the telescoping sum, and the countable Lipschitz-piece proof of the null-set property.

## References

- [1] D. H. Leung and W.-K. Tang, *Nonlinear order isomorphisms on function spaces*, Dissertationes Math. 517 (2016), 1–75; arXiv:1408.4930.
- [2] H. L. Royden and P. M. Fitzpatrick, *Real Analysis*, 4th ed., Pearson, 2010. See the Banach–Zarecki characterization of absolute continuity.